

Calvin Xu

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github://cmxu | cmxu.io | stack://cmxu

Research Statement

I am focused on making trustworthy ML techniques *usable* in the *real world*. My research lies at the intersection between *ML* and *formal methods*. I am broadly interested in the generation of practical adversarial examples, certified training, and NN verification for the vision, language, and wireless. I am currently working on LLM reasoning, agent synthesis, and coding performance/safety.

Education

Current Sep 2021	PhD Student at University of Illinois at Urbana-Champaign Working on PhD in CS, GPA: 4.0/4.0
Dec 2019 Aug 2019	Advanced Studies Student at Massachusetts Institute of Technology Coursework - Advances in Computer Vision (6.869), GPA: 5.0/5.0
Dec 2018 Aug 2015	Undergraduate/Graduate Student at Washington University in St. Louis MS in CS, Certificate in Data Mining & Machine Learning, GPA: 3.9/4.0 BS in Mathematics and BS in CS, GPA: 3.8/4.0

Publications

SEVerA: Verified Synthesis of Self-Evolving Agents

Debangshu Banerjee*, [Changming Xu](#)*, Gagandeep Singh. In Submission

Enforcing Temporal Constraints for LLM Agents

Adharsh Kamath, Sishen Zhang, [Changming Xu](#), Shubham Ugare, Gagandeep Singh, Sasa Misailovic. In Submission

Compression Aware Certified Training

[Changming Xu](#), Gagandeep Singh. Arxiv Preprint

Misaligning Reasoning with Answers—A Framework for Assessing LLM CoT Robustness

Enyi Jiang, [Changming Xu](#), Nischay Singh, Gagandeep Singh. Arxiv Preprint

Safety and Trust in Artificial Intelligence with Abstract Interpretation

Gagandeep Singh, Jacob Laurel, Sasa Misailovic, Debangshu Banerjee, Avaljot Singh, [Changming Xu](#), Shubham Ugare, Huan Zhang. FTPL

Support is all you need for Certified VAE Training

[Changming Xu](#), Debangshu Banerjee, Deepak Vasisht, Gagandeep Singh. ICLR '25

Certified DNN Training against Universal Adversarial Perturbations

[Changming Xu](#), Gagandeep Singh. ECCV '24

Scalable Relational Verification and Training for Deep Neural Networks

Debangshu Banerjee, [Changming Xu](#), Gagandeep Singh. SAIV@CAV '24

Robust Universal Adversarial Perturbations

[Changming Xu](#), Gagandeep Singh. ICML '24

Input-Relational Verification of Deep Neural Networks

Debangshu Banerjee, [Changming Xu](#), Gagandeep Singh. PLDI '24

Bypassing the Safety Training of Open-Source LLMs with Priming Attacks

Jason Vega*, Isha Chaudhary*, [Changming Xu](#)*, Gagandeep Singh. ICLR Tiny Paper '24

Exploring Practical Vulnerabilities of Machine Learning-based Wireless Systems

Zikun Liu, [Changming Xu](#), Emerson Sie, Deepak Vasisht, Gagandeep Singh. NSDI '23

Race Detection and Reachability in Nearly Series-Parallel DAGs

Kunal Agrawal, Joseph Devietti, Jeremy T. Fineman, I-Ting Angelina Lee, Robert Utterback, Changming Xu.
SODA '18

*equal contribution

Fellowships & Awards

Aug 2022	Qualcomm Innovation Fellowship Winner (one of 19 teams out of ~300) Proposal: Provably Robust Machine Learning for Wireless Systems
Aug 2019	1st place at AdvML Challenge at SigKDD 2019
Apr 2018	Dean's Select Fellowship for Research Excellence (WUSTL)
Dec 2015-17	Putnam Exam: 28, 20, 12
Aug 2015	Compton Scholar for Mathematics and Physics (4 per grade)
Apr 2017	Missouri Math Competition: 1st Place Team
Dec 2015	ICPC Regional: Top 5 Team

Work Experience

Sept 2025 May 2025	Intern at Meta LLM data generation and classification for notification ranking model
Dec 2024 Sept 2024	Intern at Bytedance Finetuning and soft prompting VLMs for content moderation
May 2021 Mar 2019	Associate Staff at MIT Lincoln Laboratory - Applied DAGAN to network traffic to augment unbalanced classes resulting in a ~20% increase in classification accuracy - Applied CNNs, Word2Vec, and Q-Learning techniques to flight data to predict flight reroutes, published at INFORMS '19 - Gathered dataset and built system using NLP and Neural Networks (NNs) to determine when to apply Adversarial ML - Explored Semantic Adversarial Examples and Spatial Transformer Networks for data augmentation - Developed an algorithm using GANs and Non-Negative NNs that won 1st place at AdvML Challenge at SigKDD 2019
Dec 2018 Sep 2016	Teaching Assistant at Washington University in St. Louis - Graded and held office hours for graduate level Machine Learning (150 students) and Numerical Applied Mathematics (50 students) - Designed, ran, and graded the final project for the Machine Learning course
Aug 2018 May 2018	Intern at EUB-INC in Beijing, China - Leveraged NNs and clustering to automate user grouping for data-driven advertisement on WeChat. - Developed algorithm which reduced turnaround time by 90% for data team, allowing for faster, more targeted advertisement

Research Experience

Current
Aug 2021

PhD Student at University of Illinois in Urbana-Champaign

Currently working on

- Certified training and efficient DNN verification of universal adversarial perturbations, VAEs, and other attacks/architectures
- Training for better LLM alignment
- Training networks that are certifiably robust under network compression (pruning/quantization)
- Probabilistic verification of VAEs in the wireless domain
- Certifiably robust training of DNNs

Mar 2018
Jun 2017

Research Assistant at Carnegie Mellon University

- Employed Adversarial ML techniques to thwart defect prediction
- Developed theory for attacking and defending high dimensional SVMs
- Implemented data poisoning attacks on the Drebin Android Malware dataset, and presented poster at CMU
- Proved that ~ 10 poisoned data points can be enough to significantly reduce the effectiveness of the malware detector

May 2017
Jun 2016

Research Assistant at Washington University in St. Louis

- Created benchmarks and proved correctness for a work-stealing scheduler, improving cache efficiency by factor of 2
- Rigorously proved a nearly series parallel race detection algorithm which matched asymptotic runtime of series parallel case
- Published paper in *ACM-SIAM SODA '18*